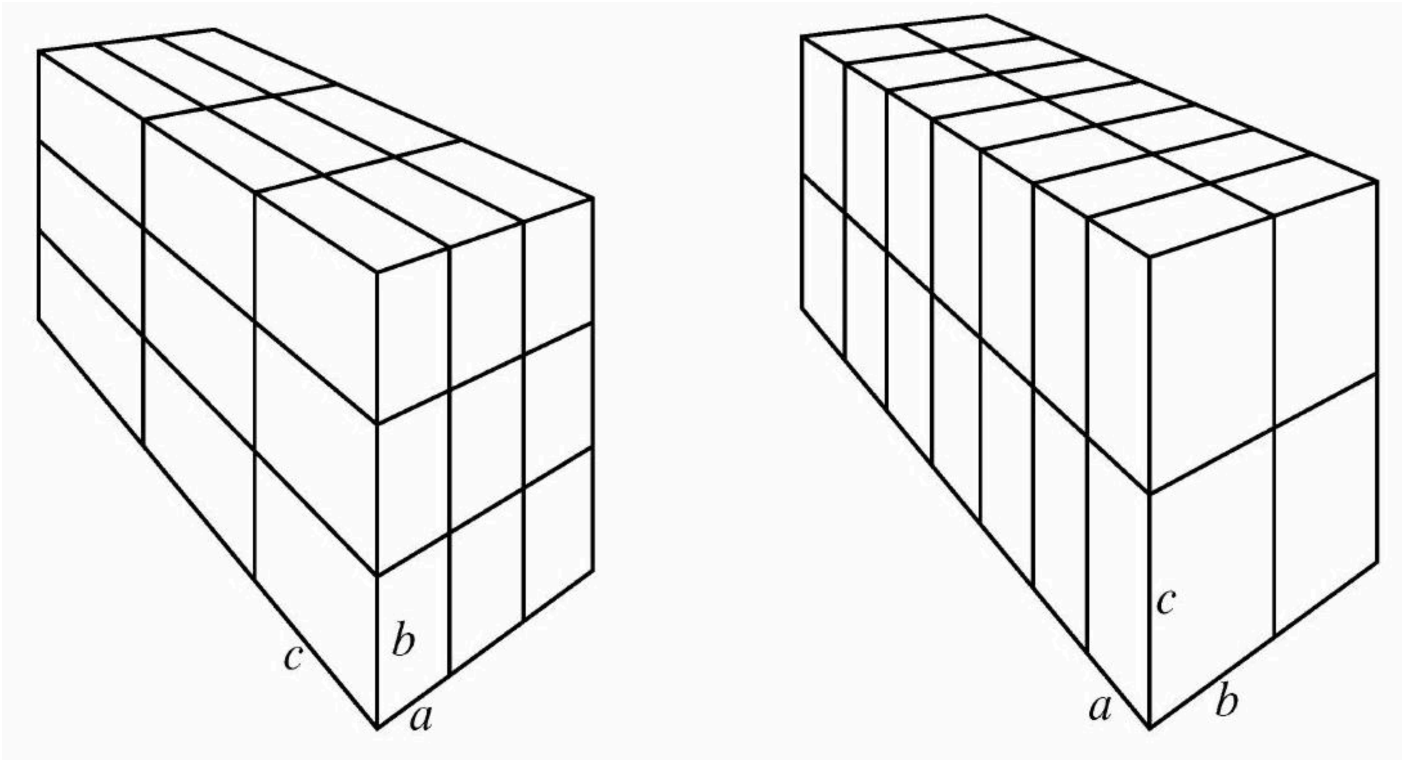
2024 AMC 10B Problem 25 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 25

Problem:

Each of 27 bricks (right rectangular prisms) has dimensions $a \times b \times c$, where a, b, and c are pairwise relatively prime positive integers. These bricks are arranged to form a $3 \times 3 \times 3$ block, as shown on the left below. A 28th brick with the same dimensions is introduced, and these bricks are reconfigured into a $2 \times 2 \times 7$ block, shown on the right. The new block is 1 unit taller, 1 unit wider, and 1 unit deeper than the old one. What is $a + b + c$?



Answer Choices:

A. 88

B. 89

C. 90

D. 91

E. 92

Solution:

Without loss of generality, assume $a < b < c$. Comparing the figures and considering the change in orientation gives rise to the equations $3a + 1 = 2b$, $3b + 1 = 2c$, and $3c + 1 = 7a$. To solve this system of linear equations, use the first two equations to write a and c in terms of b, namely $a = \frac{2}{3}b - \frac{1}{3}$ and $c = \frac{3}{2}b + \frac{1}{2}$. Substituting these into the third equation gives $\frac{9}{2}b + \frac{3}{2} + 1 = \frac{14}{3}b - \frac{7}{3}$. Multiplying both sides by 6 yields $27b + 9 + 6 = 28b - 14$, which shows that $b = 29$. Back substituting then gives $a = 19$ and $c = 44$. The requested sum is $19 + 29 + 44 = \text{(E)}\boxed{92}$.

Note: This problem was inspired by a lovely puzzle created by Thomas O'Beirn called "Melting Block".